

CE 514: Rock Mechanics

Course Code	CE 514
Course Name	Rock Mechanics
L-T-P-C	3-0-0-3
Prerequisites	Geotechnical Engineering (CE 302) or Equivalent
Intended for	UG elective and PG elective
Mutual Exclusion	None
Approval	45th BoA (Feb 2022)

Course Contents

- **Module 1:** Introduction to rock engineering, basics of geology for rock engineers, engineering and index properties of intact rocks, demonstration of laboratory techniques, design implication of different properties of intact rocks. [5 hours]
- **Module 2:** Discontinuities in rocks, engineering behaviour and characterization of discontinuities, types and description of discontinuities, orientation and spacing, discontinuity modelling, roughness, aperture, joint stiffness, RQD estimation. [5 hours]
- **Module 3:** Various geological features of rock-mass and their application in rock-mass classification (RMR, RMI, Q, GSI), correlation between different rock-mass classifications, weathering of rock-mass and its classification. [5 hours]
- **Module 4:** Deformability characteristics of jointed rock-mass, different types of moduli, challenges/issues with in-situ deformability measurement, design implications, anisotropy in rock-mass deformability, scale-effect, empirical methods, and equivalent continuum approach. [8 hours]
- **Module 5:** Shear strength characteristics of jointed rock-mass, different rock failure criteria and strength models, anisotropy in strength, scale-effect, empirical methods, equivalent continuum approach. [8 hours]
- **Module 6:** In-situ stresses in rock-mass and their importance in design of underground caverns and tunnels, permeability of rock-mass. [3 hours]
- **Module 7:** Application of rock mechanics to key rock engineering problems such as rock slope failure and stability analysis, foundations on rocks, and application of rock-mass classification in preliminary tunnel support system design. [8 hours]

Text Books

1. Aydan, Omer., **Rock Mechanics and Rock Engineering: Volume 1: Fundamentals of Rock Mechanics**, CRC Press, 2019.
2. Bieniawski, Zdzislaw T., **Design methodology in rock engineering**, CRC Press, 2020.

References:

1. Jaeger, John Conrad, Neville GW Cook, and Robert Zimmerman, **Fundamentals of rock mechanics**, John Wiley & Sons, 2009.
2. Sivakugan, Nagaratnam, Sanjay Kumar Shukla, and Braja M. Das., **Rock mechanics: an introduction**, CRC Press, 2013.
3. Goodman, Richard E., **Introduction to rock mechanics**, Vol. 2., Wiley, 1989.
4. Hudson, John A., and John P. Harrison., **Engineering rock mechanics: an introduction to the principles**, Elsevier, 2000.
5. Ramamurthy, T., ed., **Engineering in rocks for slopes, foundations, and tunnels**, PHI Learning Pvt. Ltd., 2010.
6. Pariseau, William G., **Design analysis in rock mechanics**, CRC Press, 2006.
7. Zhang, Lianyang., **Engineering properties of rocks**, Butterworth-Heinemann, 2016